
ELECTRICAL SAFETY

Fundamentals of Electrical Hazards

- Electricity is the most versatile form of energy.
- Hazards of electricity
 - misuse or failure to respect the danger
 - serious injuries, death and/or fires
- Precautions with work: design, work practices, procedures, servicing, and maintenance operations
- Inspect all electrical tools and equipment
 - prevent bodily harm, fatalities, property damage, etc.

Definitions

- **Current:** Think of current as the total volume of water flowing past a certain point in a given length of time. Electric current is measured in amperes (amps). Electric shock or injury is expressed in milli amperes (mA units or 0.001 ampere).



- **Voltage:** Think of voltage as the pressure in a pipeline. Voltage is measured in volts (v). Low voltage for this chapter is 600 v or less. Potentially hazardous voltage is between 24 v and 600 v. Potentially lethal voltage is 50 v and above (OSHA and NFPA). A car battery of 12 v direct current in a dead short can be hazardous.

Definitions (Cont.)

- **Resistance:** Think of resistance as blockage in the water pipe. Resistance is anything that retards current flow. Resistance is measured in ohms (Ω). This friction results in heat; circuits are protected by over-current devices.
- **Watt:** A watt is the quantity of electricity consumed. Consumption is measured by multiplying voltage by current ($V \times I = W$).
- **Ground:** A ground completes the electrical circuit to the earth or some conducting body to prevent electrical shock. Detect excess heat and fire protection.
- **Bonding:** Bonding is the joining of metallic parts to form an electrically conductive path. This assures electrical continuity.

Electrical Injuries

- Current flow, path, and time are the prime factors causing injuries in electrical shock.
- Severity is determined by the amount of current flowing through the victim, path through the body, and length of time the body receives the current.
- What critical parts of the body are involved?
- Is alternating current used (AC)?
- Heat-related injuries are possible.
- A person's main resistance to current flow is the skin's surface; however, a sharp decrease in resistance occurs when skin is wet or there are open wounds.

Electrical Injuries (Cont.)

- Have you ever been shocked?
 - > 3 ma = painful shock
 - > 10 ma = muscle contraction
 - > 30 ma = temporarily paralyzed lungs
 - > 50 ma = heart dysfunction (fatal)
 - 100 ma to 4 amps = fatal
 - > 4 amps = major burns and injuries

Electrical Injuries (Cont.)

- Internal Injuries
 - Electrical shock can result in chest muscle contraction leading to asphyxiation.
 - Other possible injuries include:
 - temporary paralysis
 - interference with the heart's electrical rhythm
 - severe muscular contractions
 - hemorrhages and destruction of human tissue
 - severe burns

Electrical Injuries (Cont.)

- Skin and Eye Injuries
 - tissue dies at current levels above 300 mA
 - damage to organs may not result in pain
 - thermal burns from electrical flash or arc burns
 - flashes of explosive violence
- Falls
 - shock causes muscles to contract, worker loses balance, and falls

Electrical Injuries (Cont.)

- Cardiopulmonary Resuscitation (CPR)
 - Workers on or near electrical systems must know CPR and rescue procedures.
 - Immediately start CPR on a victim of electrical shock.
 - Don't stop CPR once you start, unless a physician diagnoses death.
 - The sooner you start CPR, the better your chances are of reviving the victim.

Examples of Burns

- **Entrance Wound:** High resistance of skin transforms electrical energy into heat, which produces burns around the entrance point (dark spot in center of wound). (Source: osha.gov)



- **Exit Wound:** Current flows through the body from the entrance point, until finally exiting where the body is closest to the ground. This foot suffered massive internal injuries, which weren't readily visible, and had to be amputated a few days later. (Source: osha.gov)



Selecting Equipment

- Selection of electrical equipment
 - Make sure equipment follows recommendations of the various codes and standards.
- National Fire Protection Association (NAPA) 70 also called National Electrical Code (NEC)
- American National Standards Institute's (ANSI) C2, National Electric Safety Code
- Check state and local codes for industrial zoning requirements
- NEC code required by regulators, insurance companies, and local governments

Installing Equipment

- Always install electrical equipment in areas that are less populated.
 - If feasible, install electrical equipment in a specialized room.
- If electrical equipment is on the production floor, build protection devices around the exposed equipment (conductors, transformers, control boards, etc.).

Safety Devices

- **Interlock:** A device that interacts with another to govern succeeding operations.
 - Prevents accidental contact with hazardous parts of machine or operation (e.g., an interlocked machine guard will prevent the machine from operating unless guard is in proper place)
- **Barrier:** Prevents accidental contact with electrical equipment.
 - Dry wood and plastics have the advantage of not conducting electricity.
 - Ground all metal barriers.

Safety Devices (Cont.)

- **Warning Signs:** Display warning signs that are easy to read and grab a worker's attention near exposed current-carrying parts and in high-voltage areas.
 - Compliance with 29 CFR 1910.145.
- **Guarding:** Standard machine guarding practices can be applied to electrical equipment. Wiring provides for special hazards.
 - Ensure compliance with wiring code requirements by national and local standards.

Safety Devices (Cont.)

- **Switches:** There are several types of switches. All switches must have approved voltage and current ratings compatible with their functions.
 - knife switches, push button switches, snap switches, pendant switches, and air break switches

Protective Devices

- Safe, current-carrying capacity of conductors is determined by size, length, material, insulation, and manner of installation
- If conductors are forced to carry more than the rated safe load or heat dissipation is limited, overheating can occur.
- **Protective current devices**, such as fuses and circuit breakers, open the circuit automatically in case of excessive current flow from accidental grounds, short circuits, or overloads. Some kind of over-current device should be in every circuit.

Protective Devices (Cont.)

- **Fuses:** Link, plug, or cartridge; using the wrong kind can lead to injury. Over-fusing is a cause of overheating and may cause fires.
- **Circuit Breakers:** Are used in high-voltage circuits with large current capacities. There are two kinds:
 - **Thermal:** operates on basis of increased temperature
 - **Magnetic:** operates on amount of current that passes through the circuit
 - recommended device
 - increased temperature requires overrating circuit breaker

Protective Devices (Cont.)

- **Ground-Fault Circuit Interrupters (GFCI):** fast acting, electrical circuit-interrupting devices that are sensitive to very low levels of current flow to ground
 - designed to sense leaks of currents large enough to cause serious injury
 - operate on line-to-ground fault currents, such as insulation leakage currents, or currents likely to flow during accidental contact with a hot wire

Control Equipment

- Arrange switchboards with lockout capabilities for both AC and DC circuits
- Protect operator from live or moving parts of machinery
- Good housekeeping around the switchboard area
- Isolate switchboard in enclosed area for authorized personnel
- Use good lighting at all times
- Switch and fuse cabinets should have close-fitting doors
- Arrange connections, wiring, and equipment in an orderly manner

Control Equipment (Cont.)

- Plainly mark switches, fuses, and circuit breakers; arrange for identification of circuits and equipment
- Keep diagram or list of switchboard connections and devices posted near the equipment
- Maximize protection against accidental shock by insulating floor area within range of the live parts
- Mount motors and protect motors from dust, moisture, oils, and harmful vapors as well as misalignment, vibration, and overload
- Extension cords should be listed by UL or other recognized testing laboratory; cords should be inspected regularly and selected appropriately for function and load capacity

Test Equipment

- Test equipment regularly
- Qualified personnel should perform testing
- Examples of equipment used for testing: split-core ammeter, voltmeter, ammeter, megohmmeter, receptacle circuit tester, voltage detector, volt-ohm-milliammeter, and oscilloscopes
- Improper use of testing equipment can result in arc blast or serious injury



Specialized Processes

- High-frequency heating installations have a wide range of power capacity ranging from a few hundred watts to several hundred kilowatts and frequency ranges of 200 kilohertz (kHz) to several hundred megahertz (MHz).
- One kilowatt (kW) = one thousand watts (W)
- One kilohertz (kHz) = 1,000 hertz (Hz)
- One megahertz (MHz) = 1,000,000 (Hz)
- Burns from these processes are more painful and usually take longer to heal.

Grounding

- What is Grounding?
 - Grounding is protection from electrical shock (normally a secondary protection measure).
 - A ground is a conductive connection between the electrical circuit or equipment and the earth or ground plane. The purpose is to create a low resistance to the earth.

Grounding (Cont.)

- Codes to consider for grounding purposes
 - NFPA 70 NEC (National Electrical Code)
 - Items requiring grounding are:
 - Refrigerators, appliances using water, hand-held power tools, motor-operated appliances, any equipment in damp areas, portable hand-lamps with metallic ground guards, and some nonelectrical equipment (e.g., frames)
 - Items not requiring grounding are:
 - Approved and labeled double-insulated tools and insulated transfer tools of less than 50 v

Grounding (Cont.)

- System grounding
 - AC systems operating at 50 v or more must be grounded under a variety of voltage conditions
 - Bonding the identified conductor to a grounding electrode by means of unbroken wire
 - Ground wire insulation is usually white or gray
 - Depends on type of utility application
- Some systems are not required to be grounded
 - Some manufacturing processes can use ungrounded systems or high-impedance grounded systems
 - Highly trained personnel required
 - Can be cost effective by quick repairs, limited down time, and limited hazardous conditions

Grounding (Cont.)

- Equipment grounding:
 - Must be grounded continuously along the path
 - May be a bare conductor, the metal raceway surrounding the circuit conductors, or an insulated conductor
 - If conductor is insulated, it must have a continuous green cover or green cover with yellow stripe on it
 - Equipment-grounding conductor is always attached to the green hexagon-headed screw on receptacles, plugs, and cord connectors

Grounding (Cont.)

- Equipment grounding for fixed equipment includes noncurrent-carrying metal parts likely to become energized:
 - within 8 ft vertically or 5 ft horizontally of ground
 - located in a damp or wet location and not insulated
 - in electrical contact with metal
 - hazardous location
 - supplied by metal-clad, metal-sheathed, or metal raceway wiring method
 - operated with any terminal in excess of 150 v to ground

Grounding (Cont.)

- Equipment ground noncurrent-carrying metal parts regardless of voltage:
 - certain motor frames
 - controller cases for motors
 - electrical equipment in garages, theaters, and movie studios
 - accessible electric signs and associated equipment
 - switchboard frames and structures

Grounding (Cont.)

- Ground the following equipment:
 - frames and tracks of electrically operated cranes
 - mobile homes and recreational vehicles
 - metal enclosures around equipment carrying voltages in excess of 750 v between conductors
 - metal frames of non-electrically-driven elevator cars that have electrical conductors
 - hand-operated metal shifting ropes and cables of electric elevators

Grounding (Cont.)

- Maintenance of grounds
 - Only personnel with knowledge and training of electricity should install or repair electrical equipment.
 - Maintenance personnel should make certain that the green, insulated, equipment-grounding conductor is attached to the green hexagonal screw; and the white, grounded circuit conductor should be attached only to the silver-colored binding screw.
 - Ensure electrically continuous equipment is grounded from metal enclosure through the line cord, receptacle, and grounding system.
 - Regular maintenance and testing schedules can help predict deteriorating trends in equipment grounds.

Grounding (Cont.)

- Three-wire adapters
 - In the work place, many workers abuse items such as the three-wire adaptor by pulling out the grounding pin or cutting it off. When this is done, that operator could be holding a potentially lethal device.
- **Double-insulated tools:** Tools constructed with two separate systems of insulation reducing the chance for failure.
 - can give a false sense of security to some operators
 - best indicator for safety of a tool is the Underwriters Laboratories (UL) or recognized testing lab
 - for max protection against shock and to eliminate the need to ground the equipment, use self-contained battery-powered tools

Hazardous Locations

- **Hazardous locations:** Areas where several factors are available in combination or by themselves to allow ignition as a result of electrical causes when the following two conditions coexist:
 - The proper mix of flammable substance and oxygen are present in large enough quantities to produce an ignitable atmosphere in the area of electrical equipment.
 - An electric arc, a flame escaping from an ignited substance inside an enclosure, heat, or other source of ignition, must be present at a temperature equal to or greater than the flash point of the flammable mixture.

Hazardous Locations (Cont.)

- Hazardous locations are classified depending on the properties of the flammable vapors, liquids, gases, combustible dusts, or fibers that may be present.

Hazardous Locations (Cont.)

Class I
Vapors & Gases

Division One
Division Two

Group A–D

Class II
Combustible Dust

Division One
Division Two

Group E–G

Class III
Ignitable Flyings

Division One
Division Two

Group A: Acetylene
Group B: Hydrogen or equivalent
Group C: Ethyl-ether vapors, etc.
Group D: Gasoline, etc.

Group E: Metal dust
Group F: Carbon black, coal dust, etc.
Group G: Grain dusts

Hazardous Locations (Cont.)

- **Class I:** Flammable gases or vapors are present in the air in quantities sufficient to produce explosive or ignitable mixtures.
- **Class II:** Combustible or conductive dusts are present.
- **Class III:** Ignitable fibers are present but not likely to be in sufficient quantities to produce ignitable mixtures. (Group classifications are not applied to this class.)

Hazardous Locations (Cont.)

- **Group A:** Acetylene
- **Group B:** Hydrogen (or gases of equivalent hazard)
- **Group C:** Ethylene (or gases of equivalent hazard)
- **Group D:** Gasoline (or gases of equivalent hazard)
- **Group E:** Metal Dust
- **Group F:** Coal Dust
- **Group G:** Grain Dust

Hazardous Locations (Cont.)

- **Division 1:** The substance referred to by class is present during normal operating conditions.
- **Division 2:** The substance referred to by class is present only in abnormal conditions, such as a container failure or system breakdown.

Hazardous Locations (Cont.)

- Establishing limits
 - classify an area per NEC codes and standards for hazardous location – flammable liquids, vapors or gases, combustible dusts, and easily ignitable fibers or flyings
 - determine the degree of hazard (Division 1 or 2)
- Reducing hazards
 - remove or isolate the potential ignition source
 - control the atmosphere at the ignition source

Hazardous Locations (Cont.)

- Planning electrical installations
 - Limits of the hazardous area
 - Experience of comparable projects and understanding of specific conditions at the job site
 - Environmental aspects: prevailing winds, site topography, proximity to other structures and equipment, and climatic factors impact the extent of hazardous location
 - Factors for establishing limits: size, shape and construction features, existence of windows and doors, absence or presence of walls, enclosures, and other barriers, ventilation and exhaust systems, drainage ditches, separators, and impounding basins, quantity of hazardous materials, location of leakages, physical properties of materials, and maintenance work

Explosion-Proof Apparatus

- Defined in NEC Article 100:

Apparatus enclosed in a case capable of withstanding an explosion of a specified gas or vapor, which may occur within it, and of preventing the ignition of a specified gas or vapor surrounding the enclosure by sparks, flashes, or explosion of the gas or vapor within and which operates at such an external temperature that a surrounding flammable atmosphere will not be ignited thereby.

- Apparatus must meet requirements of the Underwriters Laboratories for use in hazardous locations.



Inspection

- Equipment should be deenergized before an inspection.
- Equipment should be considered hot until proven otherwise.
- Conduct tests on the equipment to verify that it is deenergized.
- All breakers and switches should be locked, open, grounded, and tagged out so they cannot to be reenergized until the inspection is completed.

Rotating and Intermittent-Start Equipment Inspection

- Not all machinery parts use electricity; some parts may start moving due to stored energy.
- All rotors and armatures must be blocked out before inspection is made.
- Do not wear loose clothing, wristwatches, rings, or metal pens and pencils.
- Do not use metal flashlights.

High-Voltage Equipment Inspection

- Only authorized and trained personnel should work on high-voltage equipment.
- Wear proper PPE (e.g., gloves)
 - Refer to Chapter 7; National Safety Council Occupational Safety and Health Data Sheet 12304-059, Flexible Insulating Protective Equipment for Electrical Workers; and OSHA 29 CFR 1910.132 and 1910.137, General Equipment PPE and Electrical Protective Equipment, and several other safety guideline resources (NFPA) for additional information

Link Belt Crane Accident

- What happens when the boom of a crane accidentally comes too close to a 46 kV power feeder?



Maintenance

- Only trained and experienced electricians make repairs on electrical circuits and electrical apparatus.
- Refer to NFPA 70-E for requirements for electrical maintenance.
- When dealing with electrical equipment, a good maintenance schedule is a must.
- Use only high-grade electrical equipment UL standard.
- Check equipment with testers and testing devices to see if the line is dead (fingers are not a testing device).
- Must be able to read schematic diagrams.
- Use proper PPE and always inspect/check before use as well as maintain all PPE.

Lockout / Tagout

- Make sure when purchasing electrical equipment that it has lockout / tagout capabilities.
- Every key configuration should be different.
- Color code locks.
- Tag the switch with work being done, worker's name, and the department involved.
- Follow safety-related work practices listed in 29 CFR 1910.331-339.

LOCKOUT STATION CENTRO DE SEGURIDAD



Employee Training

- Train all employees who work with hazards of electricity to read warning signs and to use guards and other protective devices and safe operational procedures.
- Never work alone with potentially hazardous electrical equipment.
- Management must develop and implement safety programs to comply with OSHA 29 CFR 1910.331-333 Safety-related work practices and power equipment or electrical energy sources.
- OSHA 29 CFR 1910.132-133 and 1910.135-138 address additional safety concerns with electrical equipment and energy sources.
- Supervisors must be kept informed of possible electrical hazards, and management must require supervision of all operations using electrical or electronic equipment.